



Machine Learning II

Data Science M.Sc.

Prof. Dr. Steffen Wagner
Berliner Hochschule für Technik

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Classification

Conditional Independence and the Naive Bayes Classifier



Recap I

A supervised machine learning method is called **regression** if the outcome variable Y is *numeric*, usually continuous.

If the outcome is a *categorical* variable then the method is called **classification**, which is what we are considering now.



Recap II: Bayes Classifier

- ▶ Suppose we have a classification problem with K possible classifications $y_1, y_2, \dots, y_K$.
- ▶ Y is our discrete or nominal outcome variable taking one of these values and depends on the predictor variables $x_1, \dots, x_p$.
- ▶ A **Bayes Classifier** is a theoretical ideal. It assigns Y to the class y_j given the values of $x_1, \dots, x_p$ according to the decision rule:
Choose the group y_j which maximises the posterior probability

$$P(Y=y_k|x_1, \dots, x_p) \quad \text{for all } k=1, \dots, K.$$



Recap III: random variabls (RV)

- ▶ The **pdf** of a continuous RV X has integral 1: $\int_{\mathbb{R}} f_X(x) dx = 1$
- ▶ and **expectation** $E(X) = \int_{\mathbb{R}} x \cdot f_X(x) dx$.
- ▶ The **expectation of a function** h of the RV X is defined as $E(h(X)) = \int_{\mathbb{R}} h(x) \cdot f_X(x) dx$.
- ▶ **example:** $h(x) = x^2 \quad \Rightarrow \quad E(X^2) = \int_{\mathbb{R}} x^2 \cdot f_X(x) dx$



Recap IV: Bayes Error Rate

- ▶ The specific probability that one observation Y_i is assigned to the wrong class is $1 - \max_k P(Y_i = y_{ik} | x_{i1}, \dots, x_{ip})$.
- ▶ The **Bayes error rate** is the integral of the specific error rates weighted by the density of X .

$$1 - \int_{\mathbb{R}} \max_k P(Y = y_k | x) f_X(x) dx.$$

- ▶ Treating $\max_k P(Y = y_k | x) := h(x)$ as a function of x this is often written as

$$1 - E \left(\max_k P(Y = y_k | X) \right).$$

This measures to what extent we can expect to make a wrong classification.



The Naive Bayes Classifier

- ▶ Under the ideal condition that the densities of X and $P(Y|X)$ are known, it can be shown that the Bayes Classifier is the best classifier we can hope to achieve, i.e. having the smallest theoretical possible error rate.
- ▶ Not only is this quite a strict condition, the Bayes classifier is not a constructive algorithm.
- ▶ **But:** In practice we can rarely obtain the unknown probability exactly, we need a method to approximate these probabilities, and there are many different methods to do this.
- ▶ In this lecture we will concentrate on an approximation called **Naive Bayes Classifier** which also serves as a baseline classification to judge more complex methods regarding their additional impact.



Bayes Theorem

Simplification:

- ▶ Let's assume that there is only one predictor X and this is a discrete random variable.
- ▶ The outcome Y variable belongs to one of K classes: $Y \in \{1, 2, \dots, K\}$.
- ▶ The classification probability simplifies to

$$P(Y=k|X=x)$$

Bayes' Theorem *reverses* the direction of the conditioning, so the expression is in terms of the probability X given Y

$$P(Y=k|X=x) = \frac{P(X=x|Y=k)p(Y=k)}{P(X=x)}.$$

- ▶ The denominator does not involve Y at all, so the value of k which maximises the left hand side also maximises $P(X=x|Y=k)P(Y=k)$.



Bayes Theorem (cont)

In other words if we compute

$$f(k) = P(X=x|Y=k)P(Y=k)$$

for every class k , then we can choose $\arg \max f(k)$ as the best class and

$$P(Y=k|X=x) = \frac{f(k)}{\sum_j f(j)}$$

In Bayesian statistics this is often express as

$$P(Y=k|X=x) \propto P(X=x|Y=k)p(Y=k)$$

with

- ▶ $P(X=x|Y=k)$: likelihood function
- ▶ $P(Y=k)$: prior probability

In many cases both the likelihood and prior are feasible to compute.



Let's complicate things (a little bit)

Now we consider the case when there are two discrete predictor variables X_1 and X_2 . Starting with $P(Y=k|X_1=x_1, X_2=x_2)$ and applying Bayes' Theorem twice, we get

$$P(Y=k|X_1=x_1, X_2=x_2) \propto P(X_1=x_1|Y=k, X_2=x_2)P(X_2=x_2|Y=k)P(Y=k).$$

Proof: homework¹

¹see https://en.wikipedia.org/wiki/Bayes%27_theorem#Bayes'_theorem_for_3_events and [https://en.wikipedia.org/wiki/Chain_rule_\(probability\)](https://en.wikipedia.org/wiki/Chain_rule_(probability))



Recap V: Independence

- ▶ Two events X_1 and X_2 are independent when

$$P(X_1|X_2) = P(X_1)$$

interpretation: "The probability of X_1 doesn't change if we know X_2 "

- ▶ Equivalently, X_1 and X_2 are independent when

$$P(X_2|X_1) = P(X_2),$$

that translates to

$$P(X_1 \cap X_2) = P(X_1)P(X_2)$$

- ▶ Example: Rolling a six and you passing the course Machine Learning II are (resp. should be²) independent events.

²only in german, but please feel free to translate and read it:

<https://taz.de/AStA-ueber-Laschets-Notenwuerfeln/!5203762/#kommune>



Conditional Independence

X_1 and X_2 are **conditionally independent given** Y when

$$P(X_1|Y, X_2) = P(X_1|Y)$$

interpretation: If we know Y then it's irrelevant what X_2 is!

Equivalently, X_1 and X_2 are conditionally independent given Y when

$$P(X_1 \cap X_2|Y) = P(X_1|Y)P(X_2|Y)$$

and

$$P(X_2|Y, X_1) = P(X_2|Y).$$



Conditional Independence: example I

The event that a BHT student chosen at random this semester ...

- ▶ studies *Statistical Computing*: X_1
- ▶ studies *Computer Science for Big Data*: X_2
- ▶ is a 1st semester Data Science Masters Student: Y

X_1 and X_2 are not independent, since $P(X_1|X_2) > P(X_1)$.

They (probably) are independent once we know whether the student is a 1st Semester Data Science student since Y “causes” X_1 and Y “causes” X_2 :

$$P(X_1|Y, X_2) = P(X_1|Y)$$



Conditional Independence: example II

The event that a person ...

- ▶ is a regular smoker: X_1
- ▶ has high blood pressure: Y
- ▶ suffers a heart attack: X_2

X_1 and X_2 are not independent but they are independent once we know that the person has high blood pressure or not since X_1 "causes" Y and Y "causes" X_2 :

$$P(X_1|Y, X_2) = P(X_1|Y) \quad \text{resp.} \quad P(X_2|Y, X_1) = P(X_2|Y)$$



Naive Bayes and Conditional Independence

In the *naive Bayes* (NB) model we assume that all of the predictor variables are conditionally independent given the class of Y .

If X_1 and X_2 are conditionally independent given Y

$$\begin{aligned} P(Y=k|X_1=x_1, X_2=x_2) &\propto P(X_1=x_1|Y=k, X_2=x_2) \cdot P(X_2=x_2|Y=k) \cdot P(Y=k) \\ &= P(X_1=x_1|Y=k) \cdot P(X_2=x_2|Y=k) \cdot P(Y=k) \end{aligned}$$



Naive Bayes and Conditional Independence (cont)

The classification having p variables and applying Bayes' Theorem p times is:

$$\begin{aligned} P(Y=k|x_1, \dots, x_p) &\propto P(x_1|Y=k, x_2, \dots, x_p) \\ &\quad \cdot P(x_2|Y=k, x_3, \dots, x_p) \\ &\quad \cdot \dots \\ &\quad \cdot P(x_p|Y=k) \\ &\quad \cdot P(Y=k). \end{aligned}$$

- ▶ **If** all of the predictor variables are conditionally independent, once the class Y is known then each of the factors $P(x_j|Y=k, x_{j+1}, \dots, x_p)$ reduces to $P(x_j|Y=k)$.
- ▶ This is called the *Naive Bayes assumption*:
 $\Rightarrow$ Knowing the class tells us all we need to know about the distribution of x_j .



Naive Bayes classification

Under the Naive Bayes assumption $P(Y=k|x_1, \dots, x_p)$ simplifies to

$$P(Y=k|x_1, \dots, x_p) \propto P(Y=k) \cdot \prod_{j=1}^p P(x_j|Y=k)$$

- ▶ Usually $P(x_j|Y=k)$ is much easier to estimate than $P(x_j|Y=k, x_{j+1}, \dots, x_p)$
- ▶ **But:** the conditional independence assumption of the Naive Bayes model is a strong assumption.



Naive Bayes Assumption

Is the strong *Naive Bayes assumption* a problem?

- ▶ On a practical level we can argue: “As long as the results are good, who cares if the predictor variables are conditionally independent on the class?”

There is a more mathematical justification *when* we are interested in classification and not in the underlying probabilities

- ▶ The conditional independence assumption may well be wrong, but the resulting difference in $P(Y=k|x_1, \dots, x_p)$ using the assumption compared to not using the assumption, is not so large.
- ▶ This difference is usually small enough, so the k which maximises the posterior probability is the same, whether or not the assumption is genuinely true.
- ▶ The *resulting classifier* will not be drastically different.



Fitting a NB classifier

To fit a NB classifier we have to estimate the terms occurring in

$$P(Y=k|x_1, \dots, x_p) \propto P(Y=k) \cdot \prod_{j=1}^p P(x_j|Y=k).$$

- ▶ $P(Y=k)$ the prior probability is easy to estimate. Just obtain the relative frequencies for Y in the data, ignoring the values of the predictor variables.
- ▶ $P(x_j|Y=k)$ is the *likelihood* of x_j given the class k . This factor will usually be fitted by assuming a distribution and estimating the parameters.



Categorical predictor variables

Special case: **Categorical predictor variables:**

- ▶ Suppose variable X_j takes values, which we can just relabel (for convenience) as "A", "B", "C",
- ▶ $P(X_j = A | Y=k)$ is estimated directly from the relative frequency of $Y=k$ from those observations where $X_j = A$, ignoring all the other $X_\bullet$ variables.

Example:

	X				
	A	B	C	D	Sum
Yes	26	27	7	4	64
No	5	5	2	0	12
Sum	31	32	9	4	76

Task: calculate the likelihoods for this example given the above definition.



Example (cont)

The likelihood terms are found by taking the relative *row* frequencies:

		X			
		A	B	C	D
	Yes	0.406	0.422	0.109	0.062
>	No	0.417	0.417	0.167	0

examples:

- ▶ $P(X_j=B|Y=Yes) = 0.422$
- ▶ $P(X_j=D|Y=No) = 0$



Continuous predictor variables

A very common assumption is to use the normal distribution density for the likelihood

$$(x_j | Y=k) \sim N(\mu_{jk}, \sigma_{jk}^2).$$

- ▶ This assumption makes the estimation of the parameters easy:
 - ▶ Estimate μ_{jk} by taking the mean of x_j for all training observations with label class k .
 - ▶ Estimate σ_{jk}^2 similarly using the observed variance of x_j in class k .
- ▶ To avoid some combinations of j and k having very small joint frequencies, the assumption is often made that the variance $\sigma_{jk}^2 = \sigma_j^2$ is independent of the class.
- ▶ Normality assumption can be released using Kernel-Density-Estimators (= Smoothers) to estimate non-parametric univariate densities (see workshop session).



Comments on NB classification

- ▶ It is a fast algorithm.
- ▶ There is no means of directly testing whether a variable improves the model or not. Splitting the data into training and test data sets is recommended to assess the accuracy of the classifier. If a model with potentially many variables is to be fitted, it is best to test each variable using cross validation.



Comparison between NB and linear discriminant analysis

- ▶ The Naive Bayes model is very similar to the linear discriminant analysis (LDA) model discussed previously.
- ▶ LDA always uses a multivariate normal distribution for the likelihood

$$P(x_1, \dots, x_p | Y=k):$$

$$(x_1, \dots, x_p | Y=k) \sim N(\mu_k, \Sigma) \quad \text{LDA,}$$

where Σ is the Variance-Covariance-Matrix, which allows the predictor variables to be correlated.

- ▶ There is no assumption of conditional independence.
- ▶ But: The predictor variables $x_1, \dots, x_p$ have to be continuous because of the normally distributed likelihood.



Comparison between NB and linear discriminant analysis (cont)

- ▶ Naive Bayes (NB) can use any distribution for the likelihood, but the correlation between x -variables is considered be zero, i.e. the variance-covariance matrix is diagonal.
- ▶ LDA is can be thought of as a “non-naive” Bayes ie (i.e. without conditional independence) with a normal likelihood.
- ▶ In both LDA and NB the common Variance-Covariance assumption between classes k can be relaxed, in discriminant analysis this results in the QDA model

$$(x_1, \dots, x_p | Y=k) \sim N(\mu_k, \Sigma_k) \quad \text{QDA.}$$